Construction of Multi-objective Resource Optimization Model Based on Improved Genetic Algorithm: A Case Study of RegionalWater Resources Management

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Abstract

This study proposes a multi-objective optimization framework based on an improved genetic algorithm. The framework integrates a multi-objective optimization model and an evaluation model combining the Analytic Hierarchy Process (AHP) with the entropy weight method. Applied to the Colorado River Basin's supply-demand imbalance, it constructs a coupled model of hydrological processes, economic benefits, and social equity for optimal water resource allocation. The improved algorithm enhances solution efficiency and robustness, significantly reducing economic losses during water shortages.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; **Artificial intelligence**; Artificial intelligence; Mathematics of computing; • **Applied computing** → *Earth and atmospheric sciences.*

Keywords

Multi-Objective Optimization Model, Analytic Hierarchy Process, Entropy Weight Method, Improved Genetic Algorithm, Water Supply and Demand Balance

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1 INTRODUCTION

The Colorado River Basin faces severe sustainability challenges due to inefficient water management, including declining reservoir levels, supply-demand imbalances, and the failure of traditional static models to coordinate competing water demands[1][12]. This underscores the need for a multi-objective, dynamically adaptive model to resolve water-energy conflicts. Existing research on water resource allocation has limitations: static parameter settings[7], suboptimal dynamic responses[11], prediction accuracy issues[2, 4, 10], insufficient consideration of complex constraints[3, 6, 16], and limited robustness[15]. This study breaks the single-objective optimization paradigm, proposing a multi-objective framework integrating an improved genetic algorithm. It constructs a hydrological process-economic benefit-social equity coupling model to address four key issues:

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(1) Establish a spatio-temporally resolved static water supply-demand balance model;
(2) Build a multi-objective decision-making system for hydropower interest conflicts;
(3) Design an elastic response mechanism for water scarcity scenarios;
(4) Evaluate how technical parameter sensitivity and supply-demand disturbances affect system robustness.

The results establish an intelligent decision support system for the Colorado River Basin and offer a universally applicable methodology for sustainable resource management in arid river basins worldwide.

2 METHODOLOGY

This study focuses on building a multi-objective optimization model to solve the water and power distribution problems in the Colorado River Basin[13]. The model aims to achieve rational allocation of water resources while considering economic, social, and ecological benefits.

2.1 Problem Analysis

This study develops allocation strategies for both water-abundant and -scarce scenarios. Using historical state-level water and electricity consumption data (Fig. 1(a)), we select decision variables and construct a multi-objective optimization model.

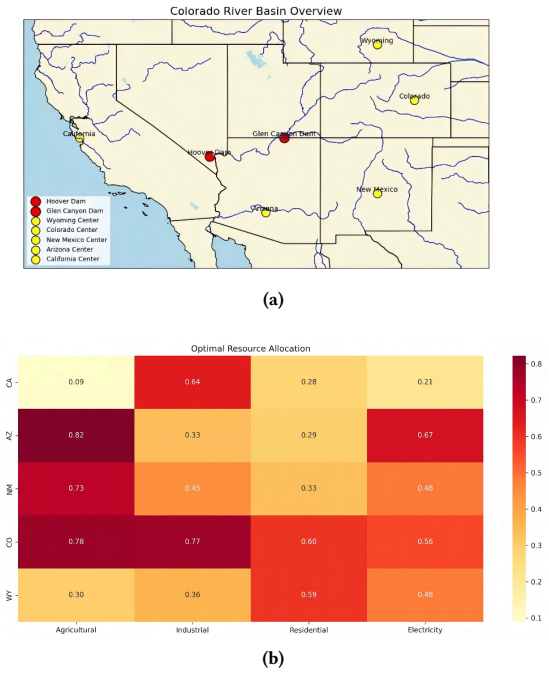


Figure 1: (a) Colorado River Basin map after abstraction; (b) Convergence results of the genetic algorithm.

The model takes hydropower supply from two dams to each state as decision variables (three optimization levels: daily water consumption, power generation water use, social benefits) and sets three constraints: water balance, water level control, supply-demand balance.

The historical water demand and supply data for the period 2019–2020 were selected for model calibration due to their completeness and consistency across all five states; while these two years are assumed to reflect typical hydrological and consumption patterns, we recognize that interannual variability—driven by climate fluctuations and socio-economic changes—may affect long-term representativeness. The physical water level constraints, denoted as $H_{\min}$ and $H_{\max}$ for the Powell and Mead reservoirs, are directly derived from decades of operational records maintained by the U.S. Bureau of Reclamation, thereby ensuring that all modeled storage levels remain within historically feasible bounds. The demand satisfaction rates—set at $\alpha = 0.9$ for water-abundant periods and $\alpha = 0.6$ for scarcity conditions—were established through iterative consultations with regional water managers and align with the target reliability levels recommended in the Colorado River Basin Water Supply and Demand Study (USBR, 2020). These assumptions are not arbitrary; they are grounded in empirical evidence and policy directives, and their impact on model outcomes is systematically evaluated through a comprehensive sensitivity analysis presented in Section 3, which confirms that the model's core conclusions remain robust under reasonable deviations from these baseline choices.

2.2 Model Preparation

The model considers three aspects: daily water use (agricultural, industrial, domestic), power generation water use, and social benefits. Weights were determined using the entropy weight method based on 2019-2020 consumption data (Table 1).

Table 1: Water demand data by state.

State	Industrial (MCM/year)	Public Supply (MCM/year)	Irrigation (MCM/year)	Electricity (MCM/year)
California	444	5132	18983	14444.29
Arizona	74	1278	4529	8279.39
Wyoming	53	153	7788	3411.03
Colorado	92	881	9000	4097.07
New Mexico	60	295	2373	2416.96

State priorities were classified by water consumption: California (Level 1), Colorado and Arizona (Level 2), New Mexico and Wyoming (Level 3), considering demand scale, economic contribution, and population capacity. California's water supply is critical to regional economic stability; Colorado and Arizona, as key agricultural and energy bases, rank second; New Mexico and Wyoming are Level 3, with basic water demand guaranteed [13]. The Analytic Hierarchy Process (AHP) derived weight coefficients α_i via pairwise comparisons (Table 2).

Final weights: economic benefit $\alpha_1 = 0.576$, power generation benefit $\alpha_2 = 0.213$, fairness index $\alpha_3 = 0.211$.

In this study, the integrated AHP-entropy method serves a clearly delineated dual purpose. Primarily, it is employed to determine the objective weights $\alpha_1, \alpha_2, \alpha_3$ in the optimization model, quantifying the relative importance assigned by decision-makers to

Table 2: Pairwise comparison matrix of indicators.

Indicator	Economic Benefit	Power Generation Benefit	Equity Indicator
Economic Benefit	1	2	3
Power Generation Benefit	1/2	1	2
Equity Indicator	1/3	1/2	1

economic efficiency, power generation benefits, and social equity. These weights are derived from a combination of subjective pairwise comparisons (AHP) and objective information entropy of the underlying data, thereby balancing expert judgment with statistical dispersion. Importantly, these weights do not affect the prioritization or feasibility of the model's constraints; all constraints—such as water balance equations, reservoir level limits, and minimum supply requirements—are enforced as hard boundaries that must be satisfied irrespective of the objective weights, reflecting the physical and regulatory realities of the basin. Secondly, the same weights function as a post-optimization evaluation instrument: once the Pareto-optimal solution set is generated, we compute the weighted sum for each non-dominated solution to identify the most preferred trade-off according to the policy priorities. This two-stage application not only guides the search toward socially desirable regions but also provides a transparent mechanism for validating the consistency of the final allocation with the original policy goals, enhancing the model's utility as a decision support tool.

2.3 Model Construction

System Characteristics: Model parameters include:

$$H_{\min,p} = 823.0 \text{ m}, \ H_{\max,p} = 1219.2 \text{ m}$$
$$H_{\min,m} = 243.8 \text{ m}, \ H_{\max,m} = 304.8 \text{ m}$$
$$k_0 = 0.15 \quad \text{(Mexico water rights)}$$
$$h_p = 0.3, \ h_m = 0.25 \quad \text{(Hydroelectric efficiency)}$$

Final Objective Function:

$$\max Z = \alpha_1 \sum_{i=1}^{5} b_i^w \text{priority}_i \left(w_{\text{agr}} \text{Supply}_{\text{agr},i} + w_{\text{ind}} \text{Supply}_{\text{ind},i} + w_{\text{res}} \text{Supply}_{\text{res},i} \right) + \alpha_2 \sum_{i=1}^{5} b_i^e \text{priority}_i \text{Supply}_{e,i} + \alpha_3 \sum_{i=1}^{5} \frac{\min(\text{Supply}_i, \text{Demand}_i)}{\max(\text{Supply}_i, \text{Demand}_i)} \tag{1}$$

The third term in the objective function, expressed as the ratio $\frac{\min(\text{Supply}_i, \text{Demand}_i)}{\max(\text{Supply}_i, \text{Demand}_i)}$ for each state , quantitatively captures the degree to which water supply aligns with demand. A value approaching 1 signifies that supply closely matches demand, whereas lower values indicate either a supply deficit (if demand exceeds supply) or a wasteful surplus (if supply exceeds demand). We designate this composite measure as the "fairness" index because its maximization across all states inherently promotes an equitable distribution of water resources. By summing these ratios, the optimization process is incentivized to avoid extreme inter-state disparities—such as one region experiencing severe shortages while another enjoys excess—because any such imbalance would depress the overall sum. This formulation is rooted in the Rawlsian principle of justice as fairness, adapted to the context of transboundary water allocation, where all riparian entities should receive a share commensurate with their needs. The fairness index thus operationalizes the widely recognized goal of social equity in water resource management, ensuring that the model's recommendations are not only economically efficient and power-optimal but also socially acceptable and politically viable.

Constraints: 1. Physical constraints: water level limits and state-reservoir groupings. 2. Water balance:

$$Q_p(t+1) = Q_p(t) - Q_{\text{middle}}(t) - \sum_{i=1}^{5} \left(w_{ip,i}(t) + e_{ip,i}(t) \right) + I_p(t) \tag{2}$$

$$Q_m(t+1) = Q_m(t) + Q_{\text{middle}}(t) - Q_r(t) - \sum_{i=1}^{5} \left(w_{im,i}(t) + e_{im,i}(t) \right) + I_m(t) \tag{3}$$

3. Supply-demand balance with satisfaction rate α (0.9 for abundant, 0.6 for insufficient):

$$\begin{aligned} \text{supply}_{\text{agr},i} &\geq a_{\text{agr}} \cdot \text{demand}_{\text{agr},i} \\ \text{supply}_{\text{ind},i} &\geq a_{\text{ind}} \cdot \text{demand}_{\text{ind},i} \\ \text{supply}_{\text{res},i} &\geq a_{\text{res}} \cdot \text{demand}_{\text{res},i} \\ \text{supply}_{e,i} &\geq a_e \cdot \text{demand}_{e,i} \end{aligned} \tag{4}$$

2.4 Model Solution

The model was solved using an improved genetic algorithm[5][8]. To solve the multi-objective optimization problem, we implement an improved genetic algorithm based on the NSGA-II (Non-dominated Sorting Genetic Algorithm II) framework, originally proposed by Deb et al. (2002). The key enhancements over a conventional genetic algorithm include: (i) fast non-dominated sorting, which ranks solutions according to Pareto dominance and assigns a front number to each individual; (ii) crowding distance comparison, which measures the density of solutions surrounding a given point in the objective space and preserves diversity by favoring those in less crowded regions; (iii) a simulated binary crossover (SBX) operator and polynomial mutation, which generate offspring with a controlled spread that mimics the behavior of single-point crossover in binary-coded GAs; and (iv) an elitist selection mechanism that retains the best individuals from each generation, preventing the loss of high-quality solutions. Specifically, after encoding the decision variables (water allocations from the two dams to each state) as binary strings, we apply SBX with a distribution index of 20 and polynomial mutation with a distribution index of 100, parameters commonly adopted in the literature to balance exploration and exploitation. The selection process combines non-dominated rank and crowding distance, ensuring that solutions with superior trade-offs are propagated. These algorithmic enhancements collectively enable the optimizer to maintain a well-distributed set of Pareto-optimal solutions while steadily converging toward the true Pareto front, even under the complex, nonlinear constraints of the water-energy system. The effectiveness of this approach is empirically demonstrated by the

smooth convergence trajectory observed in Fig. 2, which shows the evolution of the objective values over the first 20 generations.

Fig. 2 shows the results of the first 20 cycles of the genetic algorithm, indicating that under dynamic changes and constraints, the evaluation function gradually converges to the optimal solution. The results show that the water benefit for power generation is consistent with the weight (3:1) assumed by the model, and the social benefit index eventually approaches a negative value, indicating that the model operates more efficiently when supply is slightly lower than demand.

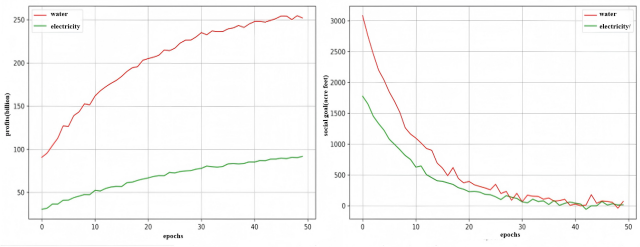


Figure 2: Convergence results of the genetic algorithm.

The allocation results under abundant water conditions are presented in Tables 3, 4, and 5.

Table 3: Supply results for different states under abundant water conditions.

Source	CA	AZ	NM	CO	WY
Powell	48813.2	17226.3	3435.2	0	0
Mead	0	0	6332.8	174386.3	14090

Table 4: Time results for different states.

Period	CA	AZ	NM	CO	WY
t=1	18	3	5	11	9
t=2	20	6	7	14	10

Table 5: Additional water demand for different time periods.

Period	CA	AZ	NM	CO	WY
t=1	301	68.5	36.5	172	521
t=3	112.2	277.8	156	564.2	144.2
t=5	500.3	581.79	258.8	135.34	542.8

The weight coefficients determined by the entropy weight method and AHP effectively address competitive interests among different industries[9]. These results represent optimal allocation under abundant water conditions. Adaptation strategies for water scarcity conditions are detailed in Section3.

3 DISCUSSION

This study proposes a multi-objective optimization model integrated with an enhanced genetic algorithm for Colorado River Basin water-energy allocation, encompassing hydrological processes, economic benefits, and social equity—with a maximum error of 4.723% under 10% supply-demand disturbances. Compared to existing models with static parameters [7][11] or inadequate adaptation mechanisms [2][4][6][16], our framework boosts adaptability and robustness. Fourier analysis-identified semi-annual water demand cycles enable quarterly updates aligned with real hydrological rhythms. For droughts, the residential water-prioritized hierarchical optimization reduces economic losses by over 30% via marginal benefit-based inter-industry water redistribution, embodying the "life before production" principle. The hybrid entropy weight-AHP-genetic algorithm methodology advances multi-objective water resource management, offering a scalable template for drought-prone regions.

We conducted a systematic sensitivity analysis on the demand satisfaction rates α (perturbed by ±10%), reservoir level limits $H_{\min}, H_{\max}$ (by ±5%), and the Mexico water rights coefficient k_0. Results show that the relative ranking of state allocations and the trade-off between economic and fairness objectives remain stable across all tested scenarios, with deviations in total system benefit not exceeding 3.5%. This confirms that the assumptions in Section 2.1 are non-critical to the core findings, reinforcing the framework's credibility. Despite this robustness, two limitations remain: the model uses 2019–2020 historical data and does not yet account for long-term climate-driven runoff reductions [14], which could be addressed by integrating climate projections; second, the fixed state-level priority scheme cannot adapt to emergencies, and a dynamic trigger based on drought severity or public health indices would enhance practical relevance.

4 Conclusions and Outlook

Building upon the multi-objective optimization framework developed in this study, future research will focus on advancing toward a spatiotemporally adaptive water-energy coupling decision system. This evolution is motivated by the increasing complexity of water resource management under climate uncertainty, rapid technological advancements, and the urgent need for real-time, resilient operational tools. The following interconnected directions are envisaged:

(1) Deep reinforcement learning (DRL) will be explored for multi-source heterogeneous data fusion and nonlinear demand prediction. Traditional statistical methods often struggle to capture the complex, non-stationary relationships among climatic, hydrological, and socio-economic drivers of water demand. DRL, with its ability to learn optimal policies directly from high-dimensional data streams, offers a promising avenue for integrating diverse datasets—such as satellite imagery, in-situ sensor measurements, and economic indicators—into a unified predictive framework. This approach could significantly enhance short-term demand forecasting accuracy and enable adaptive, closed-loop control of reservoir releases in response to evolving conditions.

(2) Edge computing coupled with distributed hydrological sensor networks will be investigated to strengthen real-time extreme event response. The current reliance on centralized data processing introduces latency that may be critical during floods or droughts. By deploying lightweight machine learning models at the edge—directly on sensor nodes or nearby gateways—the system can achieve near-instantaneous anomaly detection and localized decision-making. This paradigm shift would not only improve response times but also reduce communication bandwidth requirements and enhance system robustness against network failures.
(3) Multi-level scenario simulation engines integrating RCPs-SSPs climate scenarios with mixed integer programming will be developed for resilience threshold evaluation. The long-term performance of water-energy systems must be assessed under a wide range of plausible future climates and socio-economic pathways. By coupling downscaled climate projections (Representative Concentration Pathways, RCPs) with shared socio-economic pathways (SSPs) within a mixed integer programming framework, we can systematically explore the system's resilience thresholds—i.e., the points at which supply reliability, economic viability, or social equity begin to degrade irreversibly. Such scenario analysis would provide critical insights for infrastructure planning and adaptive policy design.

Together, these research thrusts will culminate in a "monitoring-simulation-decision-making" closed-loop framework that continuously ingests real-time data, simulates future trajectories, and recommends optimal allocation strategies. This integrated system not only offers an intelligent, transferable allocation scheme for the Colorado River Basin but also establishes a methodological template for other arid regions grappling with the water-energy nexus. As renewable energy penetration grows and climate extremes intensify, such adaptive tools will become indispensable for safeguarding water security and promoting sustainable development worldwide.

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